Project Report

Title: Basic Calendar Display

Submitted by: [Shivasharan Bhoomi Gangadhar]
Student Roll No: [54]
Student Enrollment ID: [2403111045]
Submitted to: Dr. R. S. Khamitkar
Department: Electronics and Computer Engineering
Date: 17/04/2026

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1. Project Overview

This project is a simple Python GUI application that shows the full calendar of a year. The user enters a year, and the program displays all 12 months of that year.

It uses the **calendar module** to create the calendar and **Tkinter** to design the interface. Each month is shown in a separate box with different colors, making it easy to read and understand.

This project helps to learn basic Python concepts like taking input, using functions, and creating a simple graphical interface.

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2. Problem Statement

Many applications require organized date and time representation for planning and scheduling purposes. For example, event management systems, academic planners, and personal productivity tools rely on calendars to display dates in a structured format. Manually generating a full-year calendar can be time-consuming and prone to errors.

This project aims to develop a Python-based graphical user interface (GUI) application that accepts a year as input and automatically displays the complete calendar for all 12 months in a visually organized and user-friendly format.

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3. Technology Stack

- **Programming Language:** Python 3.
- **IDE:** Visual Studio Code (VS Code).
- **Libraries Used:**
- Tkinter (for GUI development)
- Calendar (for generating calendar data)

4.Implementation

The implementation follows a simple logic:

1. 1. The user enters a **year** in the input box.
2. The program checks if the input is a **valid number**.
3. It uses the **calendar module** to create calendars for all months.
4. The old calendar on the screen is **cleared**.
5. It creates and shows all **12 months one by one**.
6. The months are arranged in a **grid format (3 rows and 4 columns)**.
7. Different **colors are used** to make the calendar look nice.

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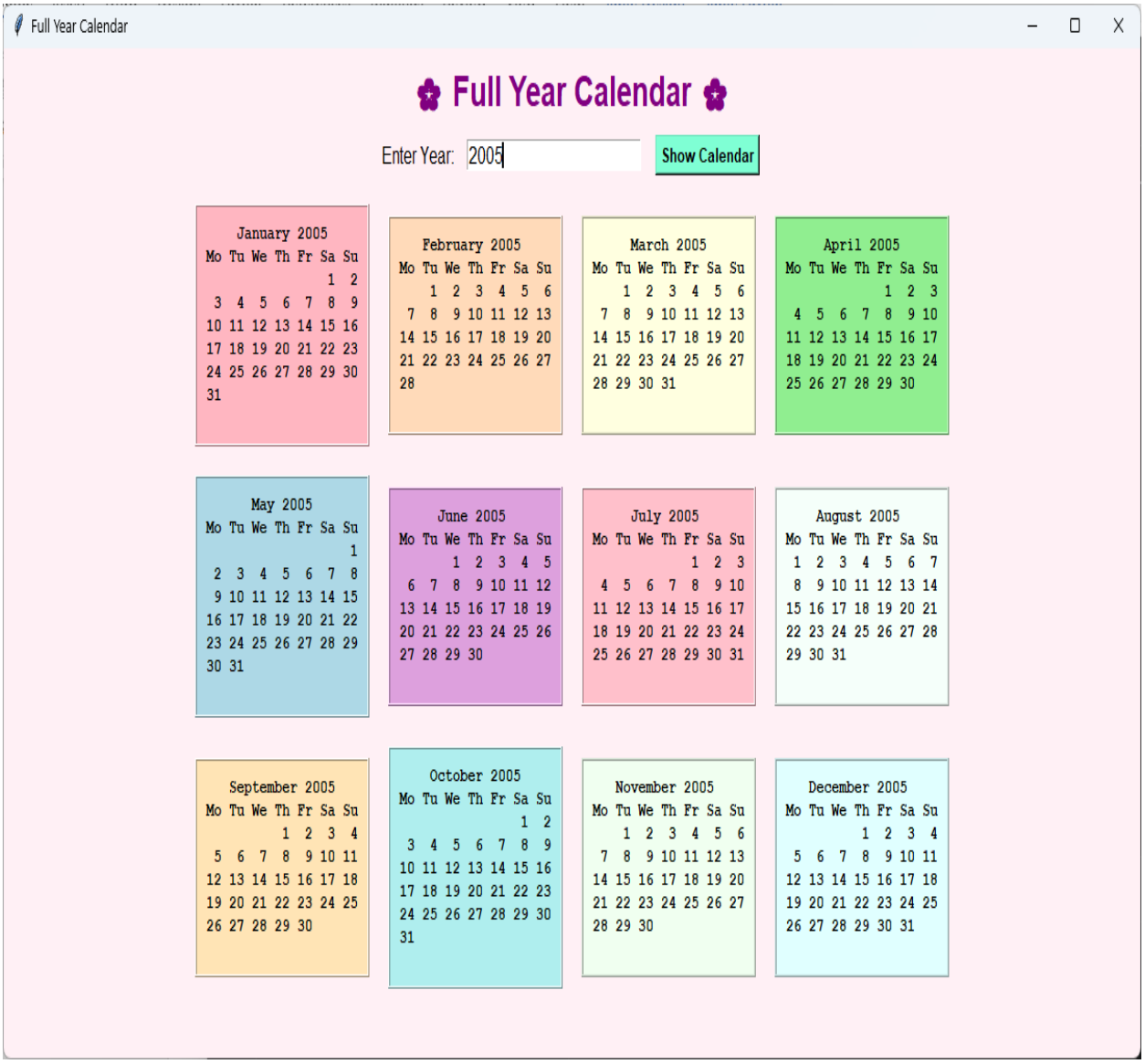
5. Code Snippet

```
python
def show_calendar():
    year = entry.get()

    # Validate 4-digit year
    if not (year.isdigit() and len(year) == 4 and int(year) > 0):
        error_label.config(text="Enter valid 4-digit year")
        return
    else:
        error_label.config(text="")

    year = int(year)

    # Clear old calendar
    for widget in calendar_frame.winfo_children():
        widget.destroy()

    # Generate months
    for month in range(1, 13):
        cal = calendar.month(year, month)

        label = tk.Label(calendar_frame, text=cal)

        row = (month - 1) // 4
        col = (month - 1) % 4

        label.grid(row=row, column=col)
```

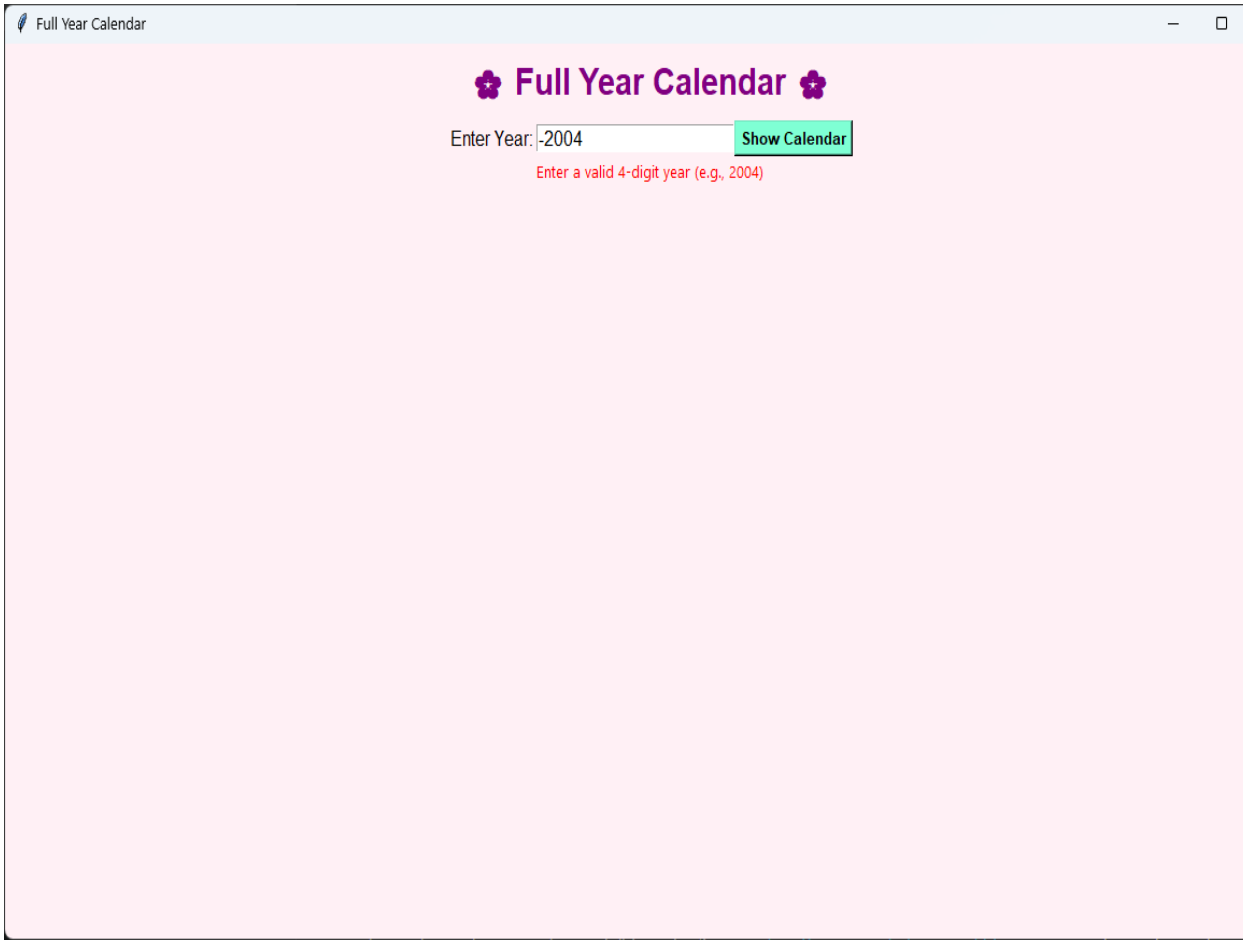
6. Sample Test Cases

Input : 2005
Output : Displays full calendar for year 2005.



6. Sample Test Cases

Input : -2004
Output : Error: Enter valid 4-digit year



7. Challenges Faced

Since this is a simple program, the challenges were minimal. However, a few difficulties were encountered:

- Handling **invalid inputs** like text or empty values required proper validation.
- Ensuring the program accepts only **4-digit valid years**.
- Managing the **GUI layout** to display all 12 months neatly in grid format.
- Adjusting **colors and design** to make the interface look clean and readable.

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8. Results and Observations

The program successfully generates and displays the full calendar for a given year. It correctly handles valid 4-digit year inputs and displays all 12 months in a structured grid format.

The program also handles invalid inputs by showing an error message, ensuring smooth user interaction. The use of colors and GUI layout improves readability and makes the calendar visually appealing.

Project Submission

Repository Link:

https://github.com/bhoomishivasharan2703-web/Basic-Calendar-Display

9. Signatures

Student Signature: _____

Guide Signature: _____

Date: _____